

Project Planning Phase

Introduction

Project Planning is one of the most important phases of the Software Development Life Cycle (SDLC). This phase defines the project scope, objectives, resources, timeline, risks, and deliverables required for successful project completion. The Library Book Request Management System is developed using the ServiceNow platform to automate the process of student book requests, validation, librarian notification, and request tracking. The planning phase ensures that the project is completed within the specified time while meeting all business requirements.

Project Objectives

The primary objectives of this project are:

Business Objectives

- Automate the library book request management process
- Reduce manual, paper-based or informal request handling
- Improve tracking and visibility of request status
- Provide students with a clear way to track their requests
- Reduce administrative workload for the librarian

Technical Objectives

- Develop the application using ServiceNow Studio
- Create custom tables and forms
- Implement business rule validation (date checks, auto-numbering)
- Generate automated email notifications using Flow Designer
- Provide reporting and dashboard visualization

Project Scope

The following functionalities are included:

Student Request Submission

Students can submit:

- New Book Requests
- Book requests with specified Issue Date and Return Date

Validation Management

- Issue date validation (cannot be in the past)
- Automatic request number generation

Status Management

- Status tracking: Pending, Approved, Rejected
- Default status assignment on submission

Notifications

- Automated email notification to librarian on new request
- Request detail summary in notification

Data Management

- Request storage in custom table
- Student information management (name, email)
- Book information management (title, author)

Reporting

- Status-based reporting (Pie Chart)
- Full request list reporting
- Dashboard for librarian overview

Out of Scope

- Integration with actual library inventory/catalog system
- Real-time book availability checking
- Fine or penalty tracking for late returns
- Mobile application version

Project Milestones

Milestone 1 — Foundation Setup

- Custom table u_library_book_request created under a new application scope
- Application menu and module configured (Library Management)
- Admin and User roles created
- Role-based permissions configured (Admin: full access, User: Create/Read/Write only)

Milestone 2 — Form & Field Configuration

- All 9 fields added to the table (Student Name, Student Email, Book Title, Author, Issue Date, Return Date, Request Number, Status, Librarian Notes)
- Form layout configured in a logical order (Student Info → Book Info → Tracking Info)
- Status choice list configured (Pending, Approved, Rejected)
- Default value set so new requests automatically start as "Pending"

Milestone 3 — Business Rules Implementation

- Business Rule created to auto-generate a unique Request Number on record creation
- Business Rule created to validate that the Issue Date is not in the past
- Both rules tested with valid and invalid data to confirm correct behavior

Milestone 4 — Notification Automation

- Flow created in Flow Designer named "Library Book Request Notification"
- Trigger configured to fire on Record Created
- Send Email action configured with dynamic data pills (Request Number, Student details, Book details)
- Flow activated and tested — execution confirmed as "Completed"

Milestone 5 — Reporting, Dashboard & Final Testing

- Pie Chart report created showing requests grouped by Status
- List report created showing all requests with key columns
- Dashboard built combining both reports for a single-screen view
- End-to-end testing performed: form submission, date validation, auto-numbering, status updates, and notification trigger all verified working